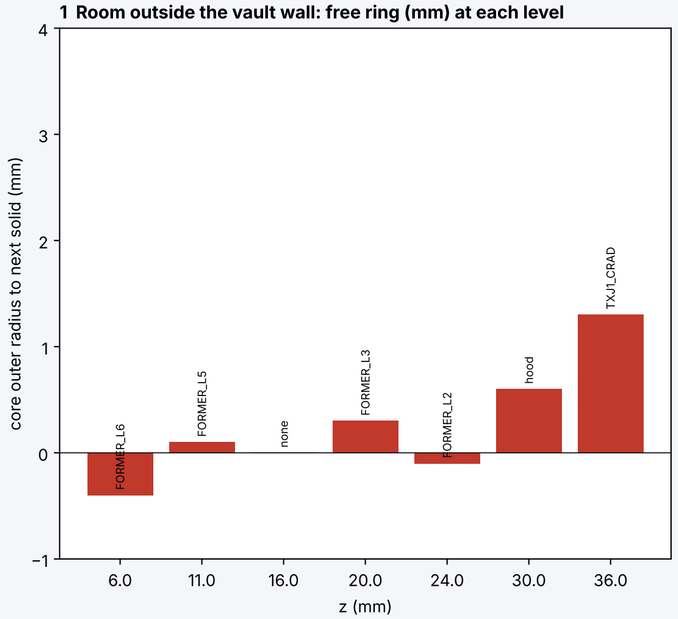
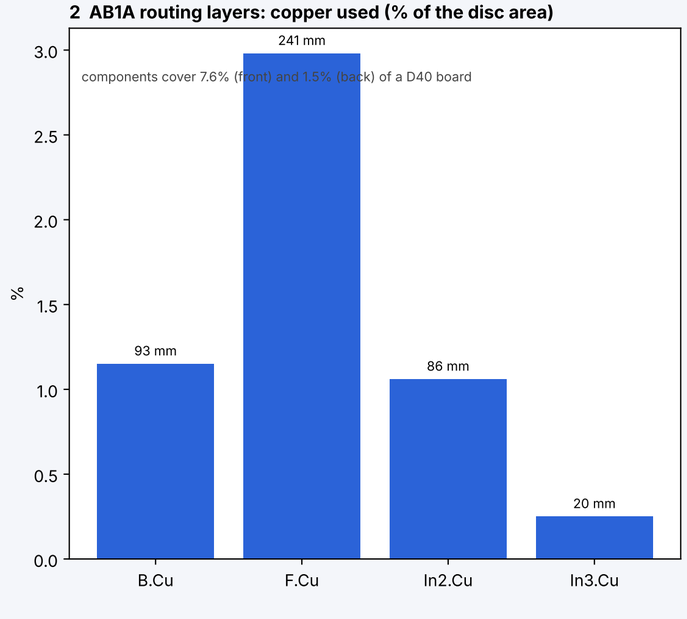
Receive-stage options model: which rule changes and additions help (MODELED, tool-measured, nothing built)

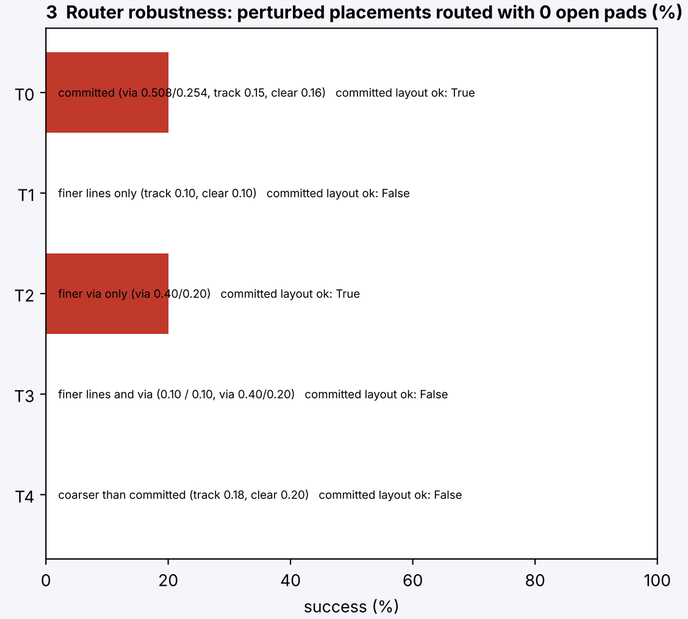
1 Room outside the vault wall: free ring (mm) at each level



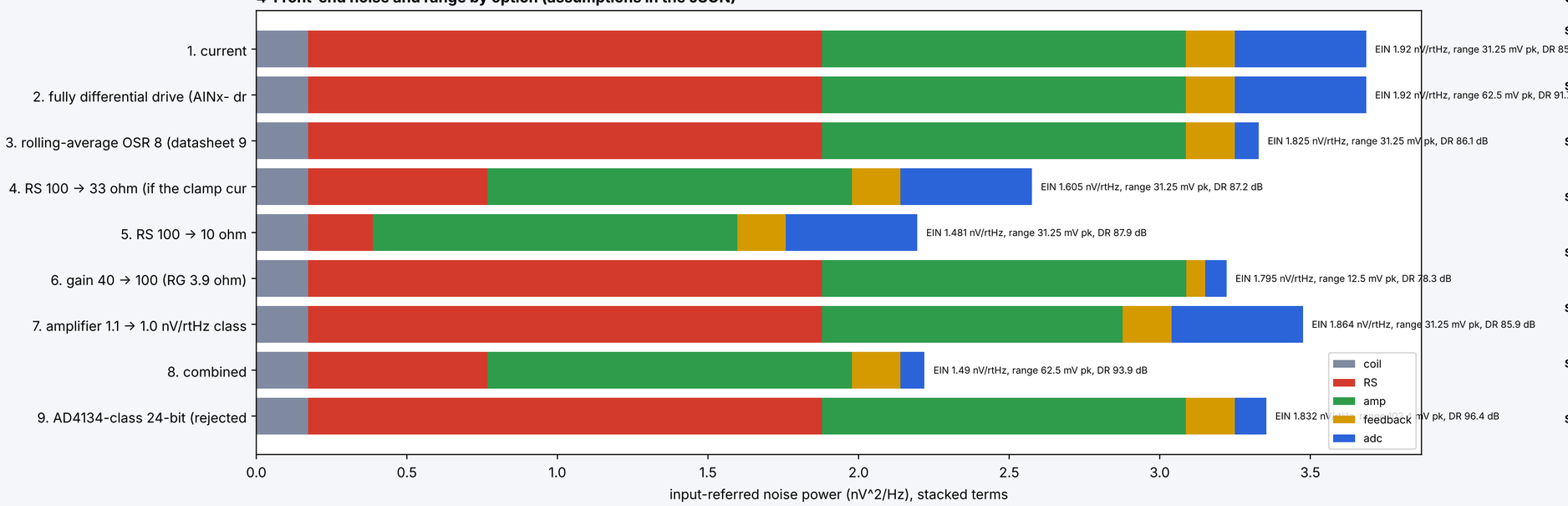
2 AB1A routing layers: copper used (% of the disc area)



3 Router robustness: perturbed placements routed with 0 open pads (%)



4 Front-end noise and range by option (assumptions in the JSON)



5 Scenarios

S0 committed design
fit: AB1 D40 pair 80.6 mm in an 88.0 mm bore (3.7 mm per side)
route 20.0% DR 85.7 dB cost: none

S1 board D44 (R_BOARD 22): no wall change
fit: two D44 need 88.6 mm: does NOT fit side by side in the 88.0 mm bore; one D44 b
route 20.0% DR 85.7 dB cost: re-place and re-route

S2 wall +1/8 inch (r 44.0 → 47.2) with D46 boards
fit: bore 94.4 mm holds two D46 (92.6 mm); area +32%
route 20.0% DR 85.7 dB cost: hub_v22 AND the six formers (their inner bore sits at r 49.1

S3 finer fab tier (0.10 / 0.10, via 0.40) with the present router
fit: no change (router-limited: 0% is the grid, see router note)
route 0.0% DR 85.7 dB cost: fabricator surcharge; needs a vendor that lists the tier (U-

S4 six to eight layers
fit: board 1.6 → about 2.0 mm (8.3 mm free above AB1)
route None% DR 85.7 dB cost: fab cost

S5 front-end changes (differential drive, RS 33, OSR 8, gain stays 40)
fit: +2 parts per channel for the differential stage
route 20.0% DR 93.9 dB cost: design time, bench; RS needs the clamp-current check; OSR ch

S6 J2 as a small board-to-board connector
fit: frees about 3 mm x 12 mm at the edge
route None% DR 85.7 dB cost: CB1D JA/JB, housing, harness (wiring_audit 53 links / 50 con

S7 combined: S2 + S5 + a rip-up router (S3 only if the fabricator needs it)
fit: area +32%, +6 to +20 dB range
route 20.0% DR 93.9 dB cost: one regeneration plus bench